Syslog framework

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Table of Figures

Table 3-1: Event context fields....10
Table 3-2: Event record fields....11
Table 3-3: Siemens SCALANCE / RUGGEDCOM validation format....11
Table 3-4: Aruba AOS-S validation format....11
Table 4-1: Required fields....13
Table 4-2: Syslog severity [1]....14
Table 4-3: event_result values....14
Table 4-4: action values....14
Table 4-5: object_type values....15
Table 4-6: Event criticality....15
Table 4-7: Conditional fields....16
Table 5-1: ISO 27001 family requirements applicable to event selection....17
Table 5-2: IEC 62443 family requirements applicable to event selection....18
Table 5-3: ISO 27001 family controls applicable to all events....20
Table 5-4: IEC 62443 family controls applicable to all events....20

Table of Contents

Table of Figures 2

Table of Contents 3

1 Framework Overview 7

1.1 Purpose 7

1.2 Scope 7

1.3 Limitations 7

2 Architecture 8

2.1 Log Sources 8

2.2 Log Transport 8

2.3 Log Collection 9

2.4 Security controls 9

3 Design Principles 10

3.1 General 10

3.1.1 Vendor neutrality 10

3.1.2 Standards alignment 10

3.1.3 Minimal required fields 10

3.1.4 Relevance 10

3.1.5 Lossless capture 10

3.2 Per-event field descriptions 10

3.3 Per-event presence in documentation validation 11

4 Event Record Schema 13

4.1 Required fields 13

4.1.1 Vocabularies 14

4.2 Conditional fields 16

5 Requirements from Standards 17

5.1 ISO 27001 Family 17

5.2 IEC 62443 Family 18

5.3 Controls applicable to all events 20

6 Event Categories 21

6.1 Access Control and Authentication 21

6.1.1 Successful Authentication (local/network) 21
6.1.2 Authentication with Elevated Privileges (local/network) 22
6.1.3 Default credentials used for login 22
6.1.4 Failed Login (local/network) 23
6.1.5 Logout 24
6.1.6 Session Timeout / Forced Logout 24
6.1.7 Remote access session initiated 25
6.1.8 Remote access session terminated 26
6.2 Identity and Account management 27
6.2.1 Account created 27
6.2.2 Account disabled 28
6.2.3 Account modified 28
6.2.4 Account Deleted 29
6.2.5 Password changed 30
6.3 Authorization and Privilege Use 31
6.3.1 Privileged command executed 31
6.3.2 Configuration mode entered 32
6.3.3 Privilege escalation 32
6.3.4 Failed Authorization attempt 33
6.3.5 Diagnostic/test interface access 34
6.4 Network-Level Access Control 34
6.4.1 Successful device identification 35
6.4.2 Unsuccessful device identification 36
6.4.3 Port-security violation 36
6.4.4 ACL deny 37
6.4.5 802.1X auth success 38
6.4.6 802.1X auth failure 38
6.5 Configuration and Change 39
6.5.1 Device configuration changed 40
6.5.2 Configuration saved or exported 40
6.5.3 Factory reset 41
6.6 System and Device State (OS / Hardware / service) 42

6.6.1 Interface status changed to UP 42
6.6.2 Interface status changed to DOWN 43
6.6.3 Service restart 43
6.6.4 System reboot 44
6.6.5 PSU failures 45
6.6.6 Fan failures 45
6.6.7 Overheating 46
6.6.8 Memory/CPU threshold exceeded 46
6.7 Network Security and Threat Detection 47
6.7.1 ARP spoofing detection 47
6.7.2 DHCP rogue server detection 48
6.7.3 STP/RSTP topology changes 49
6.7.4 VLAN hopping attempts 49
6.7.5 MAC flooding detection 50
6.8 Clock and Time Synchronization 50
6.8.1 NTP/PTP sync state – success 51
6.8.2 NTP/PTP sync state – failure 51
6.8.3 System time changed 52
6.9 Backup, Restore, Updates 53
6.9.1 Backup created 53
6.9.2 Restore initiated 54
6.9.3 Restore completed 54
6.9.4 Restore failed 55
6.9.5 Firmware deployment succeeded 55
6.9.6 Firmware deployment failed 56
6.9.7 Patch deployment succeeded 57
6.9.8 Patch deployment failed 57
6.10 Audit Log System Events 58
6.10.1 Logging started 59
6.10.2 Logging stopped 59
6.10.3 Log buffer threshold reached 60
6.10.4 Log buffer full 60

6.10.5 Log export failure 61
6.10.6 Log tampering detected 62
6.10.7 Log buffer cleared 62
6.11 Security Mechanism State 63
6.11.1 ACL enabled 63
6.11.2 ACL disabled 64
6.11.3 Port security enabled 64
6.11.4 Port security disabled 65
6.11.5 SNMP security level change 66
6.11.6 802.1X enabled 66
6.11.7 802.1X disabled 67
6.11.8 Key Management activities 67
Sources 69

1 Framework Overview

1.1 Purpose

This framework defines a vendor-neutral, standards-aligned taxonomy of security-relevant syslog events for managed OT network switches, especially the ones that are connected to Industrial Automation Control Systems (IACS). It specifies which events must be captured, the fields required to describe each event, and the criticality level assigned to each event type. The framework is intended as a reference for designing and evaluating syslog-based audit logging in OT network infrastructure.

1.2 Scope

The framework applies to managed network switches deployed in OT IACS environments and operating as syslog originators per RFC 5424 [1].

The following standards form the normative basis for event selection and field requirements:

- IEC 62443-2-1, IEC 62443-2-4, IEC 62443-3-3, IEC 62443-4-2
- ISO/IEC 27001:2022
- ISO/IEC 27002:2022
- ISO/IEC 27019:2024

The framework covers:

- Event category definitions and individual event specifications across eleven security-relevant categories.
- Required and conditional event record fields, grounded in IEC62443 CR 2.8 and ISO/IEC 27002 clause 8.15.
- Event criticality criteria with defined security levels.

The following are explicitly out of scope:

- SNMP trap handling and SNMP-based event capture
- Log normalization pipelines and vendor-specific parsing rules
- Log retention policy and storage capacity requirements
- Vendor-specific CLI configuration procedures

1.3 Limitations

Event generation is determined by device firmware and vendor implementation. This framework defines events at the semantic level – actual syslog messages produced by a given device may differ in format, field availability, and granularity. Events whose generation depends on specific device capabilities – such as IPsec support, are identified as conditional within their definitions.

The framework was validated against log output from two device lines based on their documentation. Events not observed but required by applicable standards are retained in the framework and marked as not observed. Devices that cannot forward specific security-relevant events via syslog (due to firmware limitations or hardware architecture) are outside the scope of this framework for those events. SNMP trap-based capture for such cases is identified as future work.

2 Architecture

The goal of syslog's architecture is to separate message content from message transport, while enabling easy extensibility for each layer.

The framework's architecture follows the syslog protocol model defined in RFC 5424, which separates message generation, transport, and collection into distinct functional layers. Each layer has defined responsibilities, and framework's design constraints map directly onto these layers.

2.1 Log Sources

The framework applies to managed OT network switches operating as syslog originators in the RFC 5424 model. A syslog originator is the network entity that generates syslog messages in response to local system and security events. Each switch independently generates event records and forwards them to a configured record collector.

Devices that obey RFC 5424 produce structured syslog messages with a defined header: priority value, protocol version, timestamp, hostname, application name, process ID, and message ID, with optional structured data elements. Legacy devices may produce RFC 3164-formatted messages. In such cases, timestamp precision is reduced and structured data fields are not available.

The scope of applicable log sources is limited to the management and security functions of the switch – authentication, authorization, configuration, interface states, and other event categories in Chapter 6. Traffic forwarding data and per-flow statistics are not within scope of this Framework.

2.2 Log Transport

RFC 5424 defines three transport mechanisms, differing from each other by reliability and security guarantees. The preference order for this framework is:

1) TLS transport (RFC 5425) provides message integrity, confidentiality, and mutual authentication between originator and collector. RFC 5425 specifies that TLS transport is a requirement for implementation and is recommended for general use. TLS transport is the required choice when the device capability permits so.
2) TCP transport (RFC 6587) provides ordered, reliable delivery without encryption. It is permitted where TLS is not supported by the originating device. Log messages that are transported over TCP are vulnerable to interception and modification in transit. The risk must be accepted and documented.
3) UDP transport (RFC 5426) provides no delivery guarantees, no ordering, and no connection state. It is permitted as a fallback for devices that are not capable of TCP-based transport. UDP-only devices introduce delivery and integrity risks. The absence of delivery confirmation means log loss is silent and undetectable at the transport layer.

Operators must configure each device to use the most secure transport it supports.

2.3 Log Collection

A syslog collector receives messages from one or more originators and stores them for analysis and correlation. The framework defines the semantic structure of security-relevant events so that received messages can be interpreted consistently across different devices from various vendors.

The collector implementation is out of scope of this framework. The event record schema is drafted in Chapter 4, providing a semantic reference against which received messages are assessed and categorized. Log export failure (the event when a device cannot reach its configured log collector) itself is a security-relevant event and is defined in event LOG-005.

2.4 Security controls

The syslog pipeline must be protected against compromise.

1. Transport integrity. TLS transport protects log messages in transit from modification or interception. Where TLS is unavailable, the risk of message tampering must be acknowledged. Log messages transports over unencrypted channels cannot be relied upon as tamper-evident records.
2. Collector access control. Access to the log collector and stored records must be restricted. Unrestricted access to collected logs undermines the audit trail integrity required by IEC 62443 CR 2.8.
3. Originator configuration. The syslog originator of each switch must be protected from unauthorized modification. Changes to the configuration must generate audit events. Relevant events are CFG-001 (configuration change) and, if logging is affected directly, LOG-001 and LOG-002.
4. Log capacity management. Devices with local log buffers must be configured with thresholds to prevent silent log loss, must generate alerts, as required by IEC 62443 CR 2.9. Events LOG-003 (buffer threshold reached) and LOG-004 (buffer full) in Chapter 6 address these conditions.

3 Design Principles

3.1 General

3.1.1 Vendor neutrality

Events are defined at the semantic level, independent of vendor-specific message formats, syntax, or proprietary log identifiers. An authentication failure event is defined by the security meaning, not by the character string some device emits. This generalization enables the framework to apply uniformly across the heterogeneous devices present in OT and OT IACS environments. Validation across Siemens Scalance, and Aruba devices confirms the semantic definitions hold across different vendor implementation.

3.1.2 Standards alignment

Every event category and individual event definition is derived from requirements or recommendations in IEC 62443 (2-1, 2-4, 3-3, 4-2), ISO 27001, ISO 27002, and ISO 27019. Events that do not have a traceable clause-level basis in these standards are excluded; Applicable clauses are cited at the category level in Chapter 5 and at each category header in Chapter 6.

3.1.3 Minimal required fields

The event record schema defines the minimum field set that is necessary to satisfy IEC 62443 CR 2.8 audit record requirements and ISO 27002 clause 8.15 logging requirements. Fields that are beyond this minimum are considered conditional. They are to be included only when the event type makes them applicable. This keeps the schema viable across devices with varying log outputs, without requiring field population that no device can provide.

3.1.4 Relevance

Event selection accounts for the operational characteristics of OT environments. Events that are low severity in IT contexts (such as interface state changes, system reboots, time synchronization failures) carry higher criticality in OT environments due to their direct effect on continuity of a process and audit trail integrity. The criticality scale in Table 4-6 of Chapter 4 reflects OT operational impact.

3.1.5 Lossless capture

Every event record preserves the original vendor-supplied syslog message in the “raw_message” field. No information present in the original message is discarded through extraction of structured fields. The raw message functions as the authoritative source for forensic investigation and reconstruction and as the verification reference for extracted from the message fields.

3.2 Per-event field descriptions

Where the event outcome does not alter the assigned criticality, a single event ID covers all possible outcomes. The “event_result” field in the individual record distinguishes the outcome. Where the outcome changes the security significance (therefore, the criticality), separate event IDs are defined. Tables 3-1 and 3-2 introduce the structure for evaluation of events. Each event definition includes three context fields that describe its meaning and applicability.

Table 3-1: Event context fields.

Description	States what the event is at the semantic level, independent of any vendor-specific message formats.
Rationale	Identifies the security or operational reason this event must be logged.
Condition	Specifies device capability prerequisites that must be met for the event to be generated. Absent when the event applies unconditionally.

Table 3-2: Event record fields

event_id	Framework-assigned identifier that uniquely designates the event type across the categories. Prefix indicates category (e.g., AUTH for Access Control and Authentication).
type	Classifies the event within the category, derived from the “type” element required by IEC 62443 CR 2.8 for individual audit records.
action	Identifies the operation that triggered the event. It is drawn from a controlled vocabulary defined in Chapter 4 to enable consistent filtering and correlation across vendors.
object_type	Identifies the class of entity affected by the event. Provides the structural context required by IEC 62443 CR 2.8 (“source”) and ISO 27002 clause 8.15.
object_id	Identifies the specific instance of the affected entity (e.g., a particular user account, interface, or service), populated from the device’s syslog output at record time.
event_result	Records the outcome of the event using a fixed vocabulary (success, failure, state_change, alert, denied), satisfying the “event result” element required by IEC 62443 CR 2.8.
event_criticality	Assigns a security impact level (low, medium, high, critical) based on the criteria defined in Table 4-5. For paired enable/disable events, disabling a security mechanism is assigned higher criticality than enabling it.

3.3 Per-event presence in documentation validation

Each event definition in Chapter 6 includes vendor-specific validation entries. The following two tables define the structure of those entries.

Table 3-3: Siemens SCALANCE / RUGGEDCOM validation format

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	[TEXT]
Message	[TEXT]
See also	Optional field / [TEXT]
Note	Optional field / [TEXT]

Table 3-4: Aruba AOS-S validation format

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	[TEXT]
Message	[TEXT]
See also	Optional field / [TEXT]
Note	Optional field / [TEXT]

4 Event Record Schema

Each event defined in Chapter 6 is described using a fixed set of fields. This chapter defines those fields, their purpose, and the standards requirements from which they are derived. The schema provides a consistent reference for interpretation of security-relevant events across switch devices from different vendors and satisfies audit record content requirements of IEC 62443-4-2 CR 2.8 and the logging requirements of ISO/IEC 27002 clause 8.15.

The schema consists of two categories: required fields, that are present in every event record, and conditional fields, which are populated when the event type and available device data make them applicable.

4.1 Required fields

The following table lists the fields that are present in every event record.

Table 4-1: Required fields

Field	Description	Provenance	Reference
timestamp	UTC date and time of an event	syslog	IEC62443 4-2 – CR 2.8, IEC62443 3-3 – 6.13 SR 2.11, ISO27002 – 8.15
device_id	Unique identifier of the originating device	syslog	IEC62443 4-2 – CR 2.8, ISO27002 – 8.15
severity	Raw syslog severity as assigned by the device (see Table 4-2)	syslog	N/A – protocol-native
raw_message	Original vendor-supplied syslog message, preserved without modification	syslog	N/A – protocol-native
log_source	Originating entity: system process, user action, or application	standard	IEC62443 4-2 – CR 2.8, ISO27002 – 8.15
category	Event category as defined in Chapter 6	standard	IEC62443 4-2 – CR 2.8
type	Event type within its category	standard	IEC62443 4-2 – CR 2.8
event_id	Framework-defined unique identifier for the event type	standard	IEC62443 4-2 – CR 2.8
event_result	Outcome of the event (see Table 4-3)	standard	IEC62443 4-2 – CR 2.8

action	Operation that triggered the event (see Table 4-4)	framework	Framework-defined
object_type	Class of entity affected by the event (see Table 4-5)	framework	Framework-defined
object_id	Specific instance of the affected entity, populated from device output at record time	framework	Framework-defined
event_criticality	Security impact level (see Table 4-5)	framework	Framework-defined

Syslog severity levels, as defined by RFC 5424, are listed below for reference.

Table 4-2: Syslog severity [1].

Numerical	Severity	Description
0	Emergency	System is unusable
1	Alert	Action must be taken immediately
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notice	Normal but significant condition
6	Informational	Informational messages
7	Debug	Debug-level messages

4.1.1 Vocabularies

The event_result field uses the following controlled vocabulary.

Table 4-3: event_result values

Value	Used when
Success	Operation completed as intended
Failure	Operation did not complete
Denied	Operation was blocked by a policy or access control
state_change	A system or component transitioned between states
Alert	A monitoring mechanism detected an anomalous condition

The action field uses the following controlled vocabulary.

Table 4-4: action values

Value	Used when
login	User or device authentication

logout	Session ended by user
terminate	Session ended by system or administrator
create	Account or backup creation
modify	Configuration or account change
delete	Account deletion
save	Configuration commit to storage
reset	Factory default restore
enable	Security mechanism or service activated
disable	Security mechanism or service deactivated
execute	Privileged command or diagnostic tool invocation
enter	Configuration or administrative mode access
escalate	Privilege level increased during active session
initiate	Multi-step operation started
restore	Configuration or system state restored from backup
install	Firmware or patch applied
admit	Device accepted to network
deny	Device or traffic rejected at network boundary
authenticate	Port-level authentication (802.1X)
sync	Time synchronization attempt
detect	Anomaly or fault identified by monitoring
send	Log forwarding attempt
clear	Log buffer emptied
up	Interface transitioned to operational
down	Interface transitioned to non-operational
restart	Service restart, a particular service action is interrupted
reboot	Full device restart, all services and network functions interrupted.

The object_type field uses the following controlled vocabulary.

Table 4-5: object_type values

Value	Used when
account	User accounts, identities
session	Management sessions (local and remote)
command	CLI or administrative commands
mode	Device operating modes (for example, configuration mode)
interface	Physical or logical network interfaces
config	Device configuration or parameters
device	The device itself or an external device
traffic	Network traffic matched by a policy
protocol	Network protocols (STP, ARP, DHCP)
service	System service (syslog, NTP)

security_control	Security mechanisms (ACL, port security, 802.1X)
hardware	Physical components (PSU, fans, sensors)
buffer	Local log storage
log	Audit records
clock	System time source
firmware	Device firmware
software	Software patches

Event criticality is assigned according to the following criteria.

Table 4-6: Event criticality

Criticality	Criteria
Critical	Direct security violation or use of known insecure configuration, or immediate threat to device availability or physical safety in OT environments.
High	Privileged action or repeated failure pattern
Medium	Security-relevant state change
Low	Normal operational event with audit value

4.2 Conditional fields

The following fields are populated when the event type and available device data make them applicable.

Table 4-7: Conditional fields

Field	Description	Reference
user_id	If username is involved	ISO27002 – 8.15
privilege_level	Access level of the user at the time of the event	IEC 62443 CR 2.8
source_ip	IP address of the originating entity	ISO27002 – 8.15
destination_ip	IP address of the target entity	ISO27002 – 8.15
source_port	Transport-layer port of the originating entity	ISO27002 – 8.15
destination_port	Transport-layer port of the target entity	ISO27002 – 8.15
network_protocol	Protocol involved in the event	ISO27002 – 8.15
interface	Physical or logical port on the device	-
vlan	VLAN identifier relevant to the event	-
mac_address	MAC address of the entity involved in the event	-
system_activity	Human-readable description of the event provided by the device	ISO27002 – 8.15

Note: fields without a standards reference are included as OT operational context fields.

5 Requirements from Standards

5.1 ISO 27001 Family

The following table consolidates all ISO 27001 family clauses that form the normative basis for event selection.

Table 5-1: ISO 27001 family requirements applicable to event selection

ISO 27001 FAMILY		
Source	Clause	Condition / What to log
ISO27002 – 8.15	Event logs should include for each event (as applicable): user IDs; system activities; dates, times and details of relevant events (e.g. log-on and log-off); device identity, system identifier and location; network addresses and protocols.	
ISO27001 – 8.15	Logs that record activities, exceptions, faults, and other relevant event shall be produced.	
ISO27002 – 8.20	Logging and monitoring in networks should be implemented to record and detect actions that can affect, or are relevant to, information security.	
ISO27002 – 8.18	All use of utility programs that can be capable of overriding system and application controls should be logged.	If switch includes diagnostic/maintenance tools (debug, packet capture, factory reset, forced overrides)
ISO27002 – 8.34	All access for audit purposes should be monitored and logged.	
ISO27002 – 8.16	The organization should employ technologies to monitor for anomalous communications.	
ISO27002 – 8.17	Protocols such as network time protocol (NTP) or precision time protocol (PTP) should be used to keep all networked systems in synchronization with a reference clock.	
ISO27001 – 8.16	Networks, systems and applications shall be monitored for anomalous behaviour.	
ISO27002 – 8.19	An audit log of all updates to operational software should be maintained.	
ISO27002 – 8.15	Successful and rejected system access attempts should be considered for logging.	
ISO27002 – 8.15	Successful and rejected data and other resource access attempts should be considered for logging.	
ISO27002 – 8.15	Changes to system configuration should be considered for logging.	
ISO27002 – 8.15	Use of privileges should be considered for logging.	

ISO27002 – 8.15	Use of utility programs and applications should be considered for logging.	If switch includes diagnostic/maintenance tools (debug, packet capture, factory reset, forced overrides)
ISO27002 – 8.15	Files accessed and the type of access, including deletion of important data files should be considered for logging.	
ISO27002 – 8.15	Alarms raised by the access control system should be considered for logging.	
ISO27002 – 8.15	Activation and de-activation of security systems, such as anti-virus systems and intrusion detection systems should be considered for logging.	
ISO27002 – 8.15	Creation, modification or deletion of identities should be considered for logging.	
ISO27002 – 8.24	Key management related activities should be logged and audited.	If switch uses certificates/IPsec/802.1X
ISO27019 – 8.15	Control operations carried out by operating personnel can be considered for logging.	If switch changes affect ICS traffic (VLAN, interface shutdown)
ISO27019 – 8.36	Changes to the configuration of safety systems should be logged.	
ISO27019 – 8.15	Switching operations carried out by operating personnel can be considered for logging.	
ISO27019 – 8.15	Parameter or setpoint changes carried out by operating personnel can be considered for logging.	
ISO27019 – 8.15	Changes to control programs carried out by operating personnel can be considered for logging.	
ISO27002 – 8.2	Logging all privileged access to systems for audit purposes should be considered.	
ISO27002 – 8.32	The change control procedures should include maintaining records of changes that include planning and assessing the potential impact of changes considering all dependencies; authorization of changes; communicating changes to relevant interested parties; tests and acceptance of tests for the changes; implementation of changes including deployment plans; emergency and contingency considerations including fall-back procedures;	
ISO27002 – 8.8	An audit log should be kept for all steps undertaken in technical vulnerability management.	

5.2 IEC 62443 Family

The following table consolidates all IEC 62443 family clauses that form the normative basis for event selection.

Table 5-2: IEC 62443 family requirements applicable to event selection

IEC 62443 FAMILY		
Source	Clause	Condition / What to log
IEC62443 4-2 – 6.10 CR 2.8	Individual audit records of IACS components shall include: a) timestamp; b) source (originating device, software process or human user account); c) category; d) type; e) event ID; and f) event result.	
IEC62443 2-1 – 8.4.2.1	Remote access to IACS shall be logged.	If switch offers remote management access (SSH, HTTPS, SNMP, TelNet)
IEC62443 2-4 – SP.08.02 BR	Audit logs shall include all security-related events, including user activity and account management.	
IEC62443 3-3 – 5.15 SR 1.13	The control system shall provide the capability to monitor all methods of access to the control system via untrusted networks.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for access control.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for request errors.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for operating system events.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for control system events.	If switch events impact ICS traffic
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for backup and restore events.	

IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for configuration changes.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for potential reconnaissance activity.	
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to audit log events.	
IEC62443 4-2 – 10.4 CR 6.2	IACS components shall provide the capability to continuously monitor all security mechanism performance using commonly accepted security industry practices and recommendations to detect, characterize and report security breaches in a timely manner.	
IEC62443 4-2 – 6.11 CR 2.9	IACS components shall provide auditing mechanisms to reduce the likelihood of audit record storage capacity being exceeded.	If switch has local log storage
IEC62443 4-2 – 6.11 CR 2.9	IACS components shall provide the capability to issue a warning when the allocated audit record storage volume reaches a configurable percentage of maximum audit record storage capacity.	If switch has local log storage
IEC62443 3-3 – 6.4 CR 2.2	If a component supports usage through wireless interfaces it shall provide the capability to integrate into the system that supports usage monitoring according to commonly accepted industry practices.	If the switch supports wireless interfaces.
IEC62443 4-2 – 13.3 EDR 2.13, 14.3 HDR 2.13, 15.5 NDR 2.13	Attempts to access diagnostic and test interfaces shall be actively monitored and recorded in audit logs.	
IEC62443 3-3 – 6.13 SR 2.11	Audit records should have timestamps (including date and time), generated using internal system clocks.	

5.3 Controls applicable to all events

The following ISO 27001 family controls apply to every event defined in this framework.

Table 5-3: ISO 27001 family controls applicable to all events

ISO 27001 FAMILY	
Source	Clause
ISO27001 – 8.15	Logs that record activities, exceptions, faults, and other relevant event shall be produced.
ISO27002 – 8.20	Logging and monitoring in networks should be implemented to record and detect actions that can affect, or are relevant to, information security.

The following IEC 62443 family controls apply to every event defined in this framework.

Table 5-4: IEC 62443 family controls applicable to all events

IEC 62443 FAMILY	
Source	Clause
IEC62443 2-4 – SP.08.02 BR	Audit logs shall include all security-related events, including user activity and account management.

6 Event Categories

6.1 Access Control and Authentication

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.34	All access for audit purposes should be monitored and logged.
ISO27002 – 8.15	Successful and rejected system access attempts should be considered for logging.
ISO27002 – 8.2	Logging all privileged access to systems for audit purposes should be considered.
ISO27002 – 8.15	Successful and rejected data and other resource access attempts should be considered for logging.

IEC 62443 FAMILY	
Source	Clause
IEC62443 2-4 – SP.08.02 BR	Audit logs shall include all security-related events, including user activity and account management.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for access control.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for request errors.

6.1.1 Successful Authentication (local/network)

Description	User authenticated successfully via local or network access.
Rationale	Provides traceability of successful access to the device and established a baseline for user activity monitoring and session correlation. For remote access, the event co-occurs with AUTH-007, which records the remote session context.

event_id	AUTH-001
type	Authentication Success
action	login
object_type	account
object_id	<user_id>
event_result	success
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]

Model	SCALANCE XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{Protocol}: User {User name} has logged in from {IP address}.
See also	Section 2.1.1 (local logon), Tables 2-1, 2-2, 2-3; Section 2.1.5 (network login), Tables 2-12, 2-13, 2-14, 2-15; Section 2.4.1 (access granted), Tables 2-37, 2-38, 2-39

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3362
Message	User <USERNAME> logged in from <CLIENT_IP_ADDRESS> to Telnet/WebUI/SSH/Console session
See also	Events 989, 179, 3344, 636, 637

6.1.2 Authentication with Elevated Privileges (local/network)

Description	Authentication that results in access with elevated or privileged permissions
Rationale	Privileged access introduces higher risk and must be monitored explicitly to detect misuse or unauthorized administrative actions.

event_id	AUTH-002
type	Privileged Authentication
action	login
object_type	account
object_id	<user_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	Ruggedcom confd[{pid}]: audit user: {user}/{user id} assigned to groups:{role}
See also	Table 2-3
Note	Partial coverage. SCALANCE XB/XC/XM/XR models use the same login message regardless of privilege level. Only RUGGEDCOM exposes the role via “assigned to groups:{role}”, which must be parsed.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4243
Message	Moved to manager mode for the SSH/TELNET session from IP address <IP_ADDRESS>., User USER-NAME
See also	Event 179

6.1.3 Default credentials used for login

Description	Authentication attempt performed using known default or factory-set credentials.
Rationale	Use of default credentials presents a critical security weakness; it is a direct violation of secure configuration practices. It might also point out that the configuration has not been modified since installation or that it has been reset.

event_id	AUTH-003
type	Insecure Authentication
action	login
object_type	account
object_id	<default_account>
event_result	success / failure
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: Default user {user name} logged in from {ip address}.
See also	Section 2.1.2 (local default login), Table 2-4; Section 2.1.6 (network default login), Table 2-16

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	635
Message	Alternate password used to gain access.
Note	Partial coverage. Aruba AOS-S does not flag login with default/factory credentials as a distinct event. Event 635 covers alternate recovery password use only. Default credential detection would require SIEM-level correlation of login events (3362) against known default usernames.

6.1.4 Failed Login (local/network)

Description	Authentication attempt that failed due to incorrect credentials or unauthorized access.
Rationale	Failed login attempts are essentials for detecting unauthorized access attempts and potential brute-force activity.

event_id	AUTH-004

type	Authentication Failure
action	login
object_type	account
object_id	<user_id>
event_result	failure
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{Protocol}: User {User name} failed to log in from {IP address}.
See also	Section 2.1.3 (local failed logon), Tables 2-5 through 2-9; Section 2.1.7 (network failed login), Tables 2-17 through 2-23; Section 2.6 (account lockout), Tables 2-44 through 2-48; Section 2.4.2 (access denied), Tables 2-40, 2-41

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	419
Message	Invalid user name/password on Telnet/WebUI/SSH/Console session User <USERNAME> is trying to login from <CLIENT_IP_ADDRESS>
See also	Events 180, 181, 3345, 3343, 3398, 3369, 3385

6.1.5 Logout

Description	User session is terminated normally by explicit logout action.
Rationale	Provides traceability and establishes a baseline for user activity monitoring and session correlation. For remote access, the event co-occurs with AUTH-008, which records the remote session context.

event_id	AUTH-005
type	Session Termination
action	logout
object_type	session
object_id	<session_id> or <user_id>
event_result	success
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XM-400/XR-500, M-800, S615, SC-600, W-series

Message	{protocol}: User {user name} has logged out from {ip address}.
See also	Section 2.1.4 (local logout), Table 2-10; Section 2.1.8 (network logout), Tables 2-24, 2-25, 2-26

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3363
Message	User <USERNAME> logged out of Telnet/WebUI/SSH/Console session from <CLIENT_IP_ADDRESS>

6.1.6 Session Timeout / Forced Logout

Description	User session is terminated automatically due to inactivity or admin decision.
Rationale	Indicates use of session management controls and may point out unattended sessions or policy violations. For remote access, the event co-occurs with AUTH-008 which records the remote session context.

event_id	AUTH-006
type	Session Enforcement
action	terminate
object_type	session
object_id	<session_id> or <user_id>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XM-400/XR-500, M-800, S615, SC-600, W-series
Message	The session of user {user name} was closed after {time} seconds of inactivity.
See also	Section 2.13 (remote session termination), Table 2-74; Section 2.14 (concurrent session limit), Table 2-75; Section 2.9.3 (IPsec inactivity), Table 2-56; Section 2.9.4 (OpenVPN inactivity), Table 2-57

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4242
Message	User <USERNAME> logout from <IP_ADDRESS> due to inactivity timer timeout for Telnet/WebUI/SSH/CONSOLE session
See also	Events 3387, 4241, 4831

6.1.7 Remote access session initiated

Description	Remote session to the device was established via network-based access (SSH, HTTPS). This event is generated in addition to AUTH-001 for the same access attempt. AUTH-001 records the authentication outcome and AUTH-007 records the establishment of the remote session and its network context. For remote access, both events are expected to occur.
Rationale	Remote access introduces additional attack surface and must be logged to ensure traceability and compliance with IEC 62443 requirements.

event_id	AUTH-007
type	Remote Session Start
action	initiate
object_type	session
object_id	<session_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, SC-600
Message	[IKE] <{connection name}¥|{config detail}> IKE_SA {connection name}[{config detail}] established between {ip address}[{config detail}]…{ip address}[{config detail}]
See also	Section 2.1.5 (network login), Table 2-12; Section 2.8.1 (OpenVPN established), Table 2-52; Section 2.9.1 (SINEMA RC Digital Input), Table 2-54; Section 2.9.2 (SINEMA RC Wakeup SMS), Table 2-55

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3362
Message	User <USERNAME> logged in from <CLIENT_IP_ADDRESS> to Telnet/WebUI/SSH/Console session
See also	Events 5304, 636, 637
Note	Aruba does not generate a dedicated remote session initiation event separate from the login event. Event 3362 serves both AUTH-001 and AUTH-007 — source IP and protocol type provide the remote session context. Console sessions trigger AUTH-001 only.

6.1.8 Remote access session terminated

Description	Remote session to the device was ended, either normally or not.

	This event is generated in addition to AUTH-005 for the same logout attempt and in addition to AUTH-006 for the same forced logout.
Rationale	Remote access introduces additional attack surface and must be logged to ensure traceability and compliance with IEC 62443 requirements. Session lifecycle tracking enables detection of abnormal session termination patterns.

event_id	AUTH-008
type	Remote Session End
action	terminate
object_type	session
object_id	<session_id>
event_result	success / failure
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, SC-600
Message	[IKE] <{connection name}¥|{config detail}> deleting IKE_SA {connection name}[{config detail}] between {ip address}[{config detail}]…{ip address}[{config detail}]
See also	Section 2.1.8 (network logout), Table 2-24; Section 2.8.2 (OpenVPN closed), Table 2-53; Sections 2.9.3–2.9.6 (SINEMA RC termination), Tables 2-56 through 2-59; Section 2.18.1 (session closed), Tables 2-81, 2-82

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3363
Message	User <USERNAME> logged out of Telnet/WebUI/SSH/Console session from <CLIENT_IP_ADDRESS>
See also	Events 4242, 4241
Note	Aruba does not produce a separate remote session termination event. The logout/timeout events include source IP and protocol, covering both AUTH-005/AUTH-006 and AUTH-008. Local vs. remote must be inferred from the access method field.

6.2 Identity and Account management

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.15	Creation, modification or deletion of identities should be considered for logging.
ISO27002 – 8.2	Logging all privileged access to systems for audit purposes should be considered.

IEC 62443 FAMILY	
Source	Clause
IEC62443 2-4 – SP.08.02 BR	Audit logs shall include all security-related events, including user activity and account management.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for access control.

6.2.1 Account created

Description	A new user account was created on the device.
Rationale	Account creation must be logged to ensure identity traceability and unauthorized account introduction detection.

event_id	ID-001
type	Account Lifecycle
action	create
object_type	account
object_id	<user_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} created user-account {action user name}.
See also	Section 2.5.1 (user group created), Table 2-42

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3391
Message	Local user <USERNAME> is added to the group <GROUPNAME>.
Note	Partial coverage. Aruba AOS-S does not generate a distinct "account created" event. Event 3391 records group membership assignment, not account creation itself. The traditional AOS-S model uses operator/manager role passwords rather than named user accounts. Account provisioning is reflected indirectly through group addition and password set events (2714, 2716).

6.2.2 Account disabled

Description	An existing user account is disabled.
Rationale	Account disabling is a control used to revoke access. Logging it ensures accountability and traceability.

event_id	ID-002
type	Account Lifecycle
action	disable
object_type	account
object_id	<user_id>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} deleted user-account {action user name}.
See also	Section 2.5.2 (user group deleted), Table 2-43
Note	The Scalance document labels this event "Account disabled" (Section 2.3.5) but the message text describes account deletion. No separate disable-without-delete message exists.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.
Note	Aruba AOS-S does not generate a syslog event for administratively disabling a user account. Temporary lockout due to failed logins is logged (Events 3369, 3385) but maps to AUTH-004, not ID-002.

6.2.3 Account modified

Description	Details of an existing account are changed.
Rationale	Account modifications may alter access rights and must be tracked to detect unauthorized changes.

event_id	ID-003
type	Account Modification
action	modify
object_type	account

object_id	<user_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{Protocol}: User {User name} created group {Group} and assigned to role {Role}.
See also	Section 2.5.2 (user group deleted), Table 2-43
Note	Partial coverage. No dedicated "account modified" message exists. Role/group assignment changes (Sections 2.5.1, 2.5.2) are the closest equivalent.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3391
Message	Local user <USERNAME> is added to the group <GROUPNAME>.
See also	Events 3392, 3393, 3394, 3386
Note	Partial coverage. Account modification is captured indirectly through group membership changes (3391, 3392), command authorization changes within groups (3393, 3394), and password policy changes (3386).

6.2.4 Account Deleted

Description	An existing user account is deleted from the device.
Rationale	Account deletion is important for identity lifecycle completeness. It must be logged for auditability and compliance.

event_id	ID-004
type	Account Lifecycle
action	delete
object_type	account
object_id	<user_id>
event_result	success
event_criticality	high

Examples:

Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, SCALANCE XM-400/XR-500, SCALANCE M-800, SCALANCE S615, SCALANCE SC-600,

	SCALANCE W760/W720, SCALANCE W770/W730, SCALANCE W780/W740, SCALANCE W1780/W1740
Message text	{protocol}: User {user name} deleted user-account {action user name}

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} deleted user-account {action user name}.
See also	Section 2.3.5, Table 2-36
Note	The same message covers both ID-002 and ID-004. Scalance does not differentiate between disabling and deleting an account.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3392
Message	Local user <USERNAME> is deleted from the group GROUPNAME.
Note	Partial coverage. Event 3392 records removal of a user from a group, not deletion of the account itself. Full account deletion must be inferred from a sequence of 3392 events. Password reset events (2715, 2717) revoke access but are not account deletion semantically.

6.2.5 Password changed

Description	The password of an account is changed, either by the user themselves or by an administrator.
Rationale	Password changes are critical security events that may indicate normal maintenance at times but also potential account takeover.

event_id	ID-005
type	Credentials
action	modify
object_type	account
object_id	<user_id>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} changed own password.

See also	Section 2.3.3 (password of another user changed), Tables 2-33, 2-34; Section 2.3.1 (password enabled), Table 2-30; Section 2.3.2 (password disabled), Table 2-31

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4952
Message	User <USER_NAME> password is modified.
See also	Events 2714, 2715, 2716, 2717, 2710, 4947, 4948, 4949, 4950, 4951

6.3 Authorization and Privilege Use

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.18	All use of utility programs that can be capable of overriding system and application controls should be logged.
ISO27002 – 8.15	Use of utility programs and applications should be considered for logging.
ISO27002 – 8.15	Use of privileges should be considered for logging.
ISO27002 – 8.15	Successful and rejected system access attempts should be considered for logging.

IEC 62443 FAMILY	
Source	Clause
IEC62443 2-4 – SP.08.02 BR	Audit logs shall include all security-related events, including user activity and account management.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for access control.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for request errors.
IEC62443 4-2 – 13.3 EDR 2.13, 14.3 HDR 2.13, 15.5 NDR 2.13	Attempts to access diagnostic and test interfaces shall be actively monitored and recorded in audit logs.

6.3.1 Privileged command executed

Description	Execution of a command that requires elevated privileges.
Rationale	Privileged commands affect system configuration; they must be monitored for misuse.

event_id	AUTHZ-001
type	Privileged Activity

action	execute
object_type	command
object_id	<command_name>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not observed in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	992
Message	Permit Command: user <USERNAME> command <COMMAND>.
See also	Event 5765

6.3.2 Configuration mode entered

Description	User enters a configuration or admin mode enabling system changes.
Rationale	Entering configuration mode allows to modify system settings, it must be tracked for accountability.

event_id	AUTHZ-002
type	Privileged Activity
action	enter
object_type	mode
object_id	<mode_name>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not observed in [9]. SCALANCE uses WBM without a distinct CLI configuration mode concept.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4243
Message	Moved to manager mode for the SSH/TELNET session from IP address <IP_ADDRESS>., User USER-NAME
See also	Event 179

Note	Aruba AOS-S uses a two-tier privilege model (Operator/Manager). Event 4243 records entry into Manager mode. This event also serves AUTH-002 and AUTHZ-003.

6.3.3 Privilege escalation

Description	User privileges are changed during an active session (for example, user to admin).
Rationale	Privilege escalation increases risk, must be logged to detect misuse or unauthorized elevation.

event_id	AUTHZ-003
type	Privilege Change
action	escalate
object_type	account
object_id	<user_id>
event_result	success
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not observed in [9]. Role assignment occurs at login time, not during active sessions.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4243
Message	Moved to manager mode for the SSH/TELNET session from IP address <IP_ADDRESS>., User USER-NAME
Note	Event 4243 covers privilege escalation when a user moves from Operator to Manager mode during an active session. Same event as AUTH-002 and AUTHZ-002.

6.3.4 Failed Authorization attempt

Description	An unauthorized user tries to access a resource or perform an action without privilege to do so.
Rationale	Failed attempts indicate misuse, misconfiguration, or malicious intent. They must be monitored separate from authentication failures.

event_id	AUTHZ-004

type	Authorization Failure
action	execute
object_type	command / resource
object_id	<command> or <resource_name>
event_result	denied
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	audit user: {Username}/0 Provided Invalid Password
See also	Section 2.4.2, Tables 2-40, 2-41; Section 2.10.2–2.10.5, Tables 2-62 through 2-65
Note	Partial coverage. The events above cover authentication denial rather than authorization denial. SCALANCE does not distinguish between failed authentication and failed authorization.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	993
Message	Denied Command: user <USERNAME> command <COMMAND>.

6.3.5 Diagnostic/test interface access

Description	Access to a diagnostic, test, or maintenance interface was attempted.
Rationale	Diagnostic interfaces bypass normal operational controls, they may expose sensitive data or enable unauthorized modifications. IEC 62443 requires active monitoring of such access.
Condition	Required device to expose diagnostic or test interfaces that generate syslog events on access. Applicable per IEC 62443 EDR/HDR/NDR 2.13.

event_id	AUTHZ-005
type	Diagnostic Access
action	execute
object_type	interface
object_id	<interface_name> or <tool_name>
event_result	success / failure
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—

Message	Not observed in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3005
Message	Diagnostic level is set to <SUPPORT¥|STANDARD>
Note	Partial coverage. Event 3005 records diagnostic level changes, not individual accesses to diagnostic interfaces.

6.4 Network-Level Access Control

Reference:

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.16	The organization should employ technologies to monitor for anomalous communications.
ISO27001 – 8.16	Networks, systems and applications shall be monitored for anomalous behaviour.
ISO27002 – 8.15	Successful and rejected system access attempts should be considered for logging.
ISO27002 – 8.15	Alarms raised by the access control system should be considered for logging.
ISO27002 – 8.15	Successful and rejected system access attempts should be considered for logging.

IEC 62443 FAMILY	
Source	Clause
IEC62443 2-1 – 8.4.2.1	Remote access to IACS shall be logged.
IEC62443 3-3 – 6.4 CR 2.2	If a component supports usage through wireless interfaces it shall provide the capability to integrate into the system that supports usage monitoring according to commonly accepted industry practices.
IEC62443 3-3 – 5.15 SR 1.13	The control system shall provide the capability to monitor all methods of access to the control system via untrusted networks.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for request errors.

6.4.1 Successful device identification

Description	A device was successfully identified and admitted to the network.
Rationale	Device-level authentication creates trust at the network boundary. Logging successful identification enables detection of unauthorized additions.
Condition	Requires network-level authentication mechanism (as MAC authentication, 802.1X, etc.) to be configured.

event_id	NET-001
type	Device Authentication
action	admit
object_type	device
object_id	<mac_address> or <device_id>
event_result	success
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, SC-600
Message	{firewall action accept}(1) in:{network interface} out:{network interface} len:{length} s-mac:{src mac} d-mac:{dest mac} s-ip:{ip address} d-ip:{ip address} {protocol}:{src port}->{dest port}

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.
Note	Aruba AOS-S does not generate a distinct "device identified and admitted" event. Firewall-level accept logging is not applicable to AOS-S layer-2 switches. Successful network-level authentication is covered under NET-005.

6.4.2 Unsuccessful device identification

Description	A device failed network-level identification or was denied access.
Rationale	Failed device-level authentication may indicate unauthorized hardware, misconfigurations, or an attempt to introduce an adversary device into the network.
Condition	Requires network-level authentication mechanism (as MAC authentication, 802.1X, etc.) to be configured.

event_id	NET-002
type	Device Authentication Failure
action	deny
object_type	device
object_id	<mac_address> or <device_id>
event_result	denied
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]

Model	SCALANCE M-800, S615, SC-600
Message	{firewall action reject}(1) in:{network interface} out:{network interface} len:{length} s-mac:{src mac} d-mac:{dest mac} s-ip:{ip address} d-ip:{ip address} {protocol}:{src port}->{dest port}

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.
Note	Same as NET-001. Failed 802.1X or MAC-auth is covered under NET-006.

6.4.3 Port-security violation

Description	A port-security violation was detected – an unauthorized MAC address attempted to connect.
Rationale	Port-security violations indicate unauthorized device connection attempts. Logging these violations separately enables correlation between policy enforcement and threat activity.

event_id	NET-003
type	Port Security Violation
action	deny
object_type	interface
object_id	<interface_id>
event_result	denied
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not observed in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	592
Message	<MODULE>: Move <MAC-ADDRESS> to <PORT_ID> denied
See also	Events 593, 594, 595

6.4.4 ACL deny

Description	Traffic was denied by an active access control list.

Rationale	ACL denials indicate that either policy enforcement is working as intended or that there are unauthorized access attempts present. Logging individual events supports threat detection and validation of ACL policies.

event_id	NET-004
type	ACL Enforcement
action	deny
object_type	traffic
object_id	<acl_id>
event_result	denied
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, SC-600
Message	{firewall action reject}(1) in:{network interface} out:{network interface} len:{length} s-mac:{src mac} d-mac:{dest mac} s-ip:{ip address} d-ip:{ip address} {protocol}:{src port}->{dest port}
Note	The firewall REJECT event (Section 2.2.2) serves both NET-002 and NET-004. Layer-2 SCALANCE switches (XB/XC/XM/XR) do not have equivalent ACL deny logging in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	694
Message	Src IP <IP_ADDRESS> high connection rate- port <PORT_ID>
See also	Events 695, 696
Note	Partial coverage. Aruba AOS-S ACL events (Chapter 4) are primarily configuration errors, not per-packet deny logs. Connection-rate filtering events (694–696) are the closest equivalent to real-time traffic deny logging. Per-flow ACL deny logging on AOS-S requires SNMP trap configuration.

6.4.5 802.1X auth success

Description	A device or user was successfully authenticated via 802.1X.
Rationale	802.1X authentication success confirms that network access control is functioning and the connecting entity is authorized.

event_id	NET-005
type	802.1X Authentication
action	authenticate
object_type	interface

object_id	<interface_id>
event_result	success
event_criticality	low

Note: See SEC-006/SEC-007 for events recording the activation or deactivation of 802.1X on the device.

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE W760/W720, W770/W730, W780/W740, W1780/W1740
Message	{vap}: Client {SRC mac} associated successfully.
Note	Coverage limited to WLAN models. Wired 802.1X authentication success for layer-2 switches (XB/XC/XM/XR) is not present in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	427
Message	<NUMBER_OF> auth-timeouts for the last <TIME> sec.
Note	Partial coverage. The 802.1X events chapter (Events 427–431) primarily logs failures and timeouts. Successful 802.1X authentication is typically logged through the DCA or RADIUS categories.

6.4.6 802.1X auth failure

Description	A device or user failed via 802.1X.
Rationale	802.1X authentication failure indicates an unauthorized connection attempt or misconfiguration.

event_id	NET-006
type	802.1X Authentication
action	authenticate
object_type	interface
object_id	<interface_id>
event_result	failure
event_criticality	high

Note: See SEC-006/SEC-007 for events recording the activation or deactivation of 802.1X on the device.

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE W760/W720, W770/W730, W780/W740, W1780/W1740
Message	{vap}: Client {SRC mac} failed to authenticate, status {status}.

See also	Section 2.12.2, Table 2-68; Section 2.12.5, Table 2-72; Section 2.12.6, Table 2-73

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	431
Message	Authentication failed for supp <PORT_NAME>.
See also	Events 428, 429, 430

6.5 Configuration and Change

Reference:

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.19	An audit log of all updates to operational software should be maintained.
ISO27019 – 8.15	Control operations carried out by operating personnel can be considered for logging.
ISO27019 – 8.15	Parameter or setpoint changes carried out by operating personnel can be considered for logging.
ISO27019 – 8.15	Changes to control programs carried out by operating personnel can be considered for logging.
ISO27002 – 8.15	Changes to system configuration should be considered for logging.
ISO27002 – 8.32	The change control procedures should include maintaining records of changes that include planning and assessing the potential impact of changes considering all dependencies; authorization of changes; communicating changes to relevant interested parties; tests and acceptance of tests for the changes; implementation of changes including deployment plans; emergency and contingency considerations including fallback procedures;
ISO27002 – 8.18	All use of utility programs that can be capable of overriding system and application controls should be logged.
ISO27002 – 8.15	Use of utility programs and applications should be considered for logging.
ISO27002 – 8.15	Files accessed and the type of access, including deletion of important data files should be considered for logging.
ISO27019 – 8.36	Changes to the configuration of safety systems should be logged.

IEC 62443 FAMILY	
Source	Clause
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for configuration changes.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for control system events.

6.5.1 Device configuration changed

Description	The device's configuration was modified.
Rationale	Configuration changes directly affect the device behaviour. Every modification must be logged for accountability, traceability, and detection of unauthorized changes.

event_id	CFG-001
type	Configuration Change
action	modify
object_type	config
object_id	<parameter> or <config_section>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	Device configuration changed.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3125
Message	new Startup Config Seq Num = SEQ-NUMBER by UI-TYPE.
See also	Events 3003, 3004

6.5.2 Configuration saved or exported

Description	The device's configuration was saved to persistent storage or exported externally.
Rationale	Saving or exporting configuration captures a committed state. Logging this enables distinction between different changes, and detects unauthorized configuration export.

event_id	CFG-002
type	Configuration Commit
action	save
object_type	config
object_id	<destination>
event_result	success
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, W-series
Message	{protocol}: User {user name} saved file type ConfigPack
See also	Section 2.19.1, Tables 2-84, 2-85

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3125
Message	new Startup Config Seq Num = SEQ-NUMBER by UI-TYPE.
Note	Partial coverage. Event 3125 records when startup config changes (indicating a save), but does not capture configuration export to external destinations. TFTP/SFTP transfer events would provide export evidence.

6.5.3 Factory reset

Description	The device was restored to factory default settings, removing all user-defined configuration, accounts, and security controls.
Rationale	A factory reset eliminates all the security configurations in a single operation - accounts, ACLs, port security, etc. It is the most destructive administrative action available on a managed switch. Logging the event preserves evidence of when and how the device was reset.
Condition	Required device to generate a syslog event before or during the reset sequence. Devices that reset without log messages cannot produce this event.

event_id	CFG-003
type	Configuration Override
action	reset
object_type	config
object_id	factory_default
event_result	success
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	Deleted logs by restore-factory defaults issued by user {Username}
Note	The RUGGEDCOM factory reset event is captured indirectly through the audit log cleared event (Section 2.15.1). SCALANCE layer-2 switches do not have a dedicated factory reset event in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	634
Message	Front panel clear button pressed; switch reset.
See also	Event 182

6.6 System and Device State (OS / Hardware / service)

Reference:

ISO 27001 FAMILY	
Source	Clause
ISO27001 – 8.16	Networks, systems and applications shall be monitored for anomalous behaviour.
ISO27019 – 8.15	Switching operations carried out by operating personnel can be considered for logging.

IEC 62443 FAMILY	
Source	Clause
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for operating system events.
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for control system events.

6.6.1 Interface status changed to UP

Description	A network interface changed state to UP (operational).
Rationale	Interface state changes affect network connectivity and flow of traffic. Logging transitions supports monitoring and detection of unexpected link restoration.

event_id	SYS-001
type	Interface State Change
action	up
object_type	interface
object_id	<interface_id>
event_result	state_change
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—

Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	76
Message	port <PORT_ID> is now on-line

6.6.2 Interface status changed to DOWN

Description	A network interface changed state to DOWN (non-operational).
Rationale	Interface state changes affect network connectivity and flow of traffic. Logging transitions supports monitoring and detection of unexpected lin. Unexpected interface loss may indicate hardware failure, cable disconnection, or deliberate administrative action. In OT environments, interface loss can directly affect communication between continuous processes.

event_id	SYS-002
type	Interface State Change
action	down
object_type	interface
object_id	<interface_id>
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	77
Message	port <PORT_ID> is now off-line

6.6.3 Service restart

Description	A system service or process was restarted on the device.
Rationale	Service restarts may indicate recovery from an error, or indicate instability. Logging provides traceability for root cause analysis and anomaly detection.

event_id	SYS-003
type	Service State Change

action	restart
object_type	service
object_id	<service_name>
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a single generic event.
Note	Individual services log their own enable/disable events (e.g., SSH 433/434, SNTP 410/2634, DHCP relay 690/691), but no unified "service restarted" event exists.

6.6.4 System reboot

Description	The device performed a full system reboot.
Rationale	Reboots interrupt all device functions and network services. In OT environments this directly impacts availability. Every reboot must be logged to distinguish planned maintenance (they are usually scheduled weeks or even months in advance) from unexpected failures or unauthorized restarts.

event_id	SYS-004
type	System State Change
action	reboot
object_type	device
object_id	<device_id>
event_result	state_change
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3001

Message	System reboot due to Power Failure
See also	Events 3002, 3006

6.6.5 PSU failures

Description	A power supply unit reported a failure or degraded state.
Rationale	PSU failure threatens availability of the device, may lead to uncontrolled shutdowns. Early detection enables preventive actions.
Condition	Requires hardware monitoring with PSU fault detection capability.

event_id	SYS-005
type	Hardware Fault
action	detect
object_type	hardware
object_id	<psu_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	5173
Message	System power failure
See also	Events 5174, 2769, 2777, 2778, 2779, 5263

6.6.6 Fan failures

Description	A cooling fan reported a failure or degraded state.
Rationale	Fan failure leads to high temperatures of components of the device and potential device shutdown. Logging enables early detection and replacement, incident correlations with overheating events.
Condition	Requires hardware monitoring with fan fault detection capability.

event_id	SYS-006
type	Hardware Fault
action	detect
object_type	hardware

object_id	<fan_id>
event_result	alert
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	2778
Message	Internal power supply <BAY_NUMBER>: Fan <FAN_NUM_FAULTED ¥| FAN_NUM_OK>.
See also	Event 2777

6.6.7 Overheating

Description	The device reported a temperature that exceeds a defined threshold.
Rationale	Overheating may cause hardware degradation, unexpected shutdowns, unpredictable device behavior. Logging thermal alerts supports correlation with fan and PSU failures.
Condition	Requires hardware monitoring with temperature sensors and threshold-based alerting.

event_id	SYS-007
type	Hardware Fault
action	detect
object_type	hardware
object_id	<sensor_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct named event.

Note	Chassis thermal events exist within the Chassis Events chapter but no specific "temperature threshold exceeded" event with a distinct event ID was identified. Fault Events chapter may contain relevant entries.

6.6.8 Memory/CPU threshold exceeded

Description	Memory or CPU utilization exceeded a predefined threshold.
Rationale	Monitoring resource thresholds supports early detection of abnormal load. Resource exhaustion may degrade device performance, delay processing of data, indicate denial-of-service.

event_id	SYS-008
type	Resource Alert
action	detect
object_type	hardware
object_id	<resource_type>
event_result	alert
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3007
Message	message buffer allocation has reached <USAGE_PERCENTAGE>
See also	Events 3008, 3009, 3010

6.7 Network Security and Threat Detection

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.16	The organization should employ technologies to monitor for anomalous communications.
ISO27001 – 8.16	Networks, systems and applications shall be monitored for anomalous behaviour.
ISO27002 – 8.15	Alarms raised by the access control system should be considered for logging.

IEC 62443 FAMILY	
Source	Clause

IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for potential reconnaissance activity.
IEC62443 4-2 – 10.4 CR 6.2	IACS components shall provide the capability to continuously monitor all security mechanism performance using commonly accepted security industry practices and recommendations to detect, characterize and report security breaches in a timely manner.

6.7.1 ARP spoofing detection

Description	The device detected an ARP message inconsistent with known address bindings, indicating a potential spoofing attempt.
Rationale	ARP spoofing enables man-in-the-middle attacks and traffic interception. Detection logging is essential for identifying Layer 2 threats.

event_id	THREAT-001
type	Layer 2 Attack Detection
action	detect
object_type	traffic
object_id	<interface_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	911
Message	Deny ARP OPERATION MAC-ADDR,<IP_ADDRESS> port <PORT_ID>- vlan <VLAN_ID>
See also	Events 910, 912, 913, 914, 915, 916, 917, 918, 919, 920

6.7.2 DHCP rogue server detection

Description	The device detected DHCP server responses originating from an unauthorized source.
Rationale	Rogue DHCP servers can redirect traffic, assign malicious DNS servers, or support man-in-the-middle attacks. Logging enables early identification.

event_id	THREAT-002

type	Layer 2 Attack Detection
action	detect
object_type	traffic
object_id	<interface_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	854
Message	<MODULE_NAME>: Unauthorized server <DHCPv4-SERVER-IP> detected on port <PORT_ID>
See also	Events 850, 851, 852, 855, 4546

6.7.3 STP/RSTP topology changes

Description	Spanning Tree Protocol (STP) topology change was detected on the network.
Rationale	STP topology changes alter traffic forwarding paths. While sometimes legitimate, unexpected changes may indicate an attack that attempts to manipulate Layer 2 forwarding or cause instability.

event_id	THREAT-003
type	Topology Anomaly
action	detect
object_type	protocol
object_id	<interface_id>
event_result	alert
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	845

Message	Topology Change received on port %s for %s from Source MAC<MAC_ADDRESS>
See also	Events 846, 3819, 3822, 3829, 4172

6.7.4 VLAN hopping attempts

Description	Traffic attempting to cross VLAN boundaries was detected.
Rationale	VLAN hopping bypasses network segmentation, which may allow access from a less trusted zone to critical segments.
Condition	Requires vendor-specific VLAN hopping detection capability.

event_id	THREAT-004
type	Segmentation Validation
action	detect
object_type	traffic
object_id	<interface_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.
Note	Aruba AOS-S does not generate a dedicated VLAN hopping detection event. QinQ events (Chapter 86) and BPDU events (Chapter 14) may provide partial indicators.

6.7.5 MAC flooding detection

Description	Abnormal volume of MAC address entries on a port was detected, indicating potential overflow attack.
Rationale	MAC flooding attack enables traffic sniffing. Logging supports identification of attack activity.
Condition	Requires MAC address limiting or port security with flood detection to be configured.

event_id	THREAT-005

type	Layer 2 Attack Detection
action	detect
object_type	traffic
object_id	<interface_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	592
Message	<MODULE>: Move <MAC-ADDRESS> to <PORT_ID> denied
See also	Events 593, 594, 595; Events 2539, 2541 (ARP Throttle)
Note	Partial coverage. MAC Lock events detect unauthorized MAC address activity per-port, not MAC table overflow. ARP Throttle events (2539) detect excessive ARP volume. A dedicated MAC table overflow alert is not present.

6.8 Clock and Time Synchronization

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.17	Protocols such as network time protocol (NTP) or precision time protocol (PTP) should be used to keep all networked systems in synchronization with a reference clock.

IEC 62443 FAMILY	
Source	Clause
IEC62443 3-3 – 6.13 SR 2.11	Audit records should have timestamps (including date and time), generated using internal system clocks.

6.8.1 NTP/PTP sync state – success

Description	The device successfully synchronized its clock with the configured time source.
Rationale	Accurate timestamps are a prerequisite for log correlation and forensic reconstruction. Logging successful synchronization confirms time integrity.

event_id	TIME-001
type	Time Synchronization
action	sync
object_type	service
object_id	<time_source>
event_result	success
event_criticality	low

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	413
Message	Updated time by NUM seconds from server at <IP_ADDRESS>. Previous time was <TIME>. Current time is <TIME>.
See also	Events 410, 4907, 4908

6.8.2 NTP/PTP sync state – failure

Description	The device failed to synchronize its clock with the configured time source.
Rationale	Loss of time synchronization undermines the reliability of all subsequent log timestamps. It may indicate network issues, misconfigurations, or problems with the time source itself.

event_id	TIME-002
type	Time Synchronization
action	sync
object_type	service
object_id	<time_source>
event_result	failure
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	414

Message	Unable to reach configured SNTP servers
See also	Events 411, 412, 2631

6.8.3 System time changed

Description	The device's system clock was manually adjusted.
Rationale	Manual time changes can break event sequences in audit logs. Logging the change with old and new values preserves timeline integrity, supports detection of timestamp manipulation.

event_id	TIME-003
type	Time Modification
action	modify
object_type	clock
object_id	<device_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	178
Message	Updated time by <CURRENT_TIME - OLD_TIME> seconds. Previous time was <DATE_STR>. Current time is <CURRENT DATE_STR>
See also	Events 177, 4908

6.9 Backup, Restore, Updates

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.19	An audit log of all updates to operational software should be maintained.
ISO27002 – 8.8	An audit log should be kept for all steps undertaken in technical vulnerability management.
ISO27002 – 8.15	Files accessed and the type of access, including deletion of important data files should be considered for logging.

IEC 62443 FAMILY

Source	Clause
IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for backup and restore events.

6.9.1 Backup created

Description	Configuration or system backup was created on or exported from the device.
Rationale	Backup creation establishes a recovery point. Logging it ensures traceability of what was preserved and by whom, supporting detection of unauthorized backup creation.

event_id	REC-001
type	Backup
action	create
object_type	config
object_id	<backup_id> or <destination>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE M-800, S615, W-series
Message	{protocol}: User {user name} saved file type ConfigPack
See also	Section 2.19.1, Tables 2-84, 2-85

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	Aruba AOS-S does not generate a dedicated "backup created" syslog event. Configuration saves (Event 3125) and TFTP/USB file transfers (Events 796, 98) provide indirect evidence.

6.9.2 Restore initiated

Description	Configuration or system restore operation was started.
Rationale	Restore operations replace the current state of the system. Logging provides accountability and enables correlation with following success or failure events.

event_id	REC-002
type	Restore

action	initiate
object_type	config
object_id	<backup_id> or <source>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: Loaded file type Config (restart required).
See also	Section 2.20.3, Tables 2-91, 2-92, 2-93, 2-94

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	5690
Message	Configuration restore is initiated.

6.9.3 Restore completed

Description	Configuration or system restore operation completed successfully.
Rationale	Confirms that the device is now in a previous state. Required for audit trail completeness and verification that recovery procedure was successfully executed.

event_id	REC-003
type	Restore
action	restore
object_type	config
object_id	<backup_id> or <source>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} loaded file type Firmware {version} (restart required).
See also	Section 2.20.1, Tables 2-88, 2-89
Note	Section 2.20.1 is titled "Restore successfully done" but the message refers to firmware loading. Configuration restore is covered in Section 2.20.3.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	5693
Message	Delta computation between running-config and config <CONFIG_FILE_NAME> is successful.
See also	Events 5695, 5696
Note	Partial coverage. No single "restore completed successfully" event exists. Event 5693 confirms the comparison succeeded; successful application must be inferred from absence of failure events (5639, 5694, 5697).

6.9.4 Restore failed

Description	Configuration or system restore operation failed.
Rationale	When restore failed, the device is left in a potentially inconsistent state. Logging enables identification and might trigger manual recovery procedures.

event_id	REC-004
type	Restore
action	restore
object_type	config
object_id	<backup_id> or <source>
event_result	failure
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: Failed to load file type Firmware.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	5639
Message	Configuration restore to <RESTORE_FILE_NAME> has failed- started recovery to source config.
See also	Events 5694, 5697, 5636, 5637, 5638

6.9.5 Firmware deployment succeeded

Description	Firmware update was successfully applied to the device.

Rationale	Firmware changes alter the entire operational behavior of the device. Logging successful firmware deployments keeps a version history, satisfying software update audit requirements.

event_id	REC-005
type	Firmware Update
action	install
object_type	firmware
object_id	<firmware_version>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: User {user name} loaded file type Firmware {version} (restart required).
See also	Section 2.20.1, Tables 2-88, 2-89

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4244
Message	User <USERNAME>: <IMAGE STR> Image updated via <TFTP¥|SFTP> from <SERVER IP> completed. Firmware version: Before update: <OLD VERSION> After update, <NEW_VERSION>
See also	Events 93, 163

6.9.6 Firmware deployment failed

Description	Firmware update failed to apply.
Rationale	Failed firmware deployment may leave device with a vulnerable or outdated version of firmware, or in an unstable state. Logging supports root cause analysis, ensuring the failure is not ignored.

event_id	REC-006
type	Firmware Update
action	install
object_type	firmware
object_id	<firmware_version>
event_result	failure
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG, XM-400/XR-500, M-800, S615, SC-600, W-series
Message	{protocol}: Failed to load file type Firmware.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4245
Message	User <USERNAME>: <IMAGE STR> Image update failed via <TFTP¥|SFTP> from <SERVER IP> for <PLATFORM>. Firmware version: Current version: <OLD VERSION> Expected version after update, <NEW_VERSION>
See also	Events 162, 164, 165

6.9.7 Patch deployment succeeded

Description	Software patch was successfully applied to the device.
Rationale	Patches address known vulnerabilities or fix defects in the previous versions. Logging successful patch deployments keeps a patch deployment history, providing evidence and supporting compliance.

event_id	REC-007
type	Patch Update
action	install
object_type	software
object_id	<patch_id>
event_result	success
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	The other partition was imaged successfully. A reboot is required to boot the other partition.
Note	Patch events are specific to RUGGEDCOM RX1400 only (Section 2.20.4). SCALANCE switches do not have separate patch mechanisms.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.

Note	Aruba AOS-S does not differentiate patches from firmware updates. Firmware update events (4244, 4245) cover all software deployments.

6.9.8 Patch deployment failed

Description	Software patch failed to apply.
Rationale	Patches address known vulnerabilities or fix defects in the previous versions. Failed patch leaves a known vulnerability unaddressed. Logging ensures visibility and accountability.

event_id	REC-008
type	Patch Update
action	install
object_type	software
object_id	<patch_id>
event_result	failure
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	A failure was encountered in the upgrade process.
See also	Section 2.20.5, Tables 2-96 through 2-102

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	See REC-007 note.

6.10 Audit Log System Events

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.15	Files accessed and the type of access, including deletion of important data files should be considered for logging.

IEC 62443 FAMILY	
Source	Clause

IEC62443 4-2 – 6.10 CR 2.8	The components of IACS shall provide the capability to generate audit records relevant to security for audit log events.
IEC62443 4-2 – 10.4 CR 6.2	IACS components shall provide the capability to continuously monitor all security mechanism performance using commonly accepted security industry practices and recommendations to detect, characterize and report security breaches in a timely manner.
IEC62443 4-2 – 6.11 CR 2.9	IACS components shall provide auditing mechanisms to reduce the likelihood of audit record storage capacity being exceeded.
IEC62443 4-2 – 6.11 CR 2.9	IACS components shall provide the capability to issue a warning when the allocated audit record storage volume reaches a configurable percentage of maximum audit record storage capacity.

6.10.1 Logging started

Description	The device's logging service was activated/resumed.
Rationale	Logging activation explicitly marks the beginning of record generation. Logging this event establishes the earliest point from which the audit trail is in its logical flow.

event_id	LOG-001
type	Log Service State
action	enable
object_type	service
object_id	syslog
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4331
Message	syslog: Information logging started on the SYSLOG server <SERVER_ADDRESS> over <TRANSPORT_PROTOCOL> protocol

6.10.2 Logging stopped

Description	The device's logging service was deactivated/interrupted.

Rationale	If the logging is stopped, there is a blind spot created in the audit trail. Whether intentional or not, it must be recorded as the last event of the chain of events before the visibility is lost.

event_id	LOG-002
type	Log Service State
action	disable
object_type	service
object_id	syslog
event_result	state_change
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	4332
Message	syslog: Information logging stopped on the SYSLOG server <SERVER_ADDRESS> over <TRANSPORT_PROTOCOL> protocol

6.10.3 Log buffer threshold reached

Description	The local log buffer usage reached a configured warning threshold.
Rationale	Early warning of capacity limits enables preventative actions before log data is lost due to buffer overflow.
Condition	Requires device to have local log buffer storage with configurable capacity thresholds (IEC 62443 CR 2.9).

event_id	LOG-003
type	Log Capacity Warning
action	detect
object_type	buffer
object_id	<buffer_id>
event_result	alert
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—

Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.

6.10.4 Log buffer full

Description	The local log buffer is full. New events may be dropped / oldest entries may be overwritten.
Rationale	A full buffer means active loss of records. This event signals that integrity of logs can no longer be guaranteed without immediate action.
Condition	Requires device to have local log buffer storage with configurable capacity thresholds (IEC 62443 CR 2.9).

event_id	LOG-004
type	Log Capacity Exceeded
action	detect
object_type	buffer
object_id	<buffer_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9]

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed.

6.10.5 Log export failure

Description	The device failed to forward log data to the configured remote log server.
Rationale	Export failure breaks centralized log collection. Events may only exist locally, being at risk of loss. Logging the failure ensures the gap can be monitored.

event_id	LOG-005
type	Log Transport Failure

action	send
object_type	service
object_id	<syslog_server>
event_result	failure
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	469
Message	User <USER>: SSL connection failed for Syslog session from <IP_ADDRESS>
See also	Event 472
Note	Partial coverage. Events 469 and 472 cover TLS-related syslog connection failures. A generic "syslog server unreachable" event is not present.

6.10.6 Log tampering detected

Description	The device detected unauthorized actions (modification, deletion, or clearing) on audit records.
Rationale	Log tampering is strictly forbidden. It is a direct indicator of malicious activity that is aimed at concealing actions.
Condition	Requires device to have log integrity monitoring capability.

event_id	LOG-006
type	Log Integrity Violation
action	detect
object_type	log
object_id	<log_id> or <buffer_id>
event_result	alert
event_criticality	critical

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]

Event ID	—
Message	Not observed.

6.10.7 Log buffer cleared

Description	The local log buffer was cleared by administrative action.
Rationale	Clearing the log buffer removes locally stored audit records. Whether performed for maintenance or with malicious intent, this action must itself be recorded as the last entry before the buffer is emptied. Without this event, a legitimate buffer clear is indistinguishable from log tampering.
Condition	The device must support administrative clearing of the log buffer.

event_id	LOG-007
type	Log Administration
action	clear
object_type	buffer
object_id	<buffer_id>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	Deleted logs by restore-factory defaults issued by user {Username}

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	184
Message	Log cleared as a result of 'clear logging' command

6.11 Security Mechanism State

ISO 27001 FAMILY	
Source	Clause
ISO27002 – 8.15	Activation and de-activation of security systems, such as anti-virus systems and intrusion detection systems should be considered for logging.
ISO27002 – 8.24	Key management related activities should be logged and audited.

IEC 62443 FAMILY	
Source	Clause

IEC62443 4-2 – 10.4 CR 6.2	IACS components shall provide the capability to continuously monitor all security mechanism performance using commonly accepted security industry practices and recommendations to detect, characterize and report security breaches in a timely manner.

6.11.1 ACL enabled

Description	An access control list (ACL) was activated on the device.
Rationale	ACL activation changes traffic filtering on the device. Logging confirms that restrictions are in effect.

event_id	SEC-001
type	Security Mechanism State
action	enable
object_type	security_control
object_id	<acl_id>
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].
Note	Scalance firewall rule activation is managed through configuration change events (Section 2.16), not as dedicated ACL state events.

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	ACL application to ports/VLANs generates warnings only on failure (Events 43–55). Successful ACL application is not logged separately.

6.11.2 ACL disabled

Description	An access control list (ACL) was deactivated on the device.
Rationale	ACL disabling removes traffic filtering restrictions on the device, weakening the security. Logging confirms that restrictions are not in effect.

event_id	SEC-002
type	Security Mechanism State
action	disable

object_type	security_control
object_id	<acl_id>
event_result	state_change
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	ACL removal is not explicitly logged. Event 52 logs failure to delete an applied ACL, but successful removal generates no event.

6.11.3 Port security enabled

Description	Port security was activated on an interface.
Rationale	Activation of port security restricts which devices can connect to a physical port, introducing network access control.

event_id	SEC-003
type	Security Mechanism State
action	enable
object_type	security_control
object_id	<interface_id>
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	Port security configuration changes are captured under general configuration change events (Event 3125) but no dedicated "port security enabled" event exists.

6.11.4 Port security disabled

Description	Port security was deactivated on an interface.
Rationale	Disabling port security allows any device to connect to the physical port, removing restriction.

event_id	SEC-004
type	Security Mechanism State
action	disable
object_type	security_control
object_id	<interface_id>
event_result	state_change
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.

6.11.5 SNMP security level change

Description	The SNMP security configuration was modified. (State changed from SNMPv3 to SNMPv2, for example.)
Rationale	SNMP is used for device monitoring, and sometimes configuration management. Changes to SNMP security level affect the confidentiality and integrity of the management channel. Downgrading from SNMPv3 to SNMPv2 removes authentication and encryption, for example.

event_id	SEC-005
type	Security Mechanism State
action	modify
object_type	security_control
object_id	snmp
event_result	state_change
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.
Note	SNMP configuration changes are captured under general configuration events (Event 3125). No dedicated SNMP security level change event exists.

6.11.6 802.1X enabled

Description	802.1X port-based network access control was activated on the device/on a specific interface.
Rationale	Enabling 802.1X introduces authenticated access control at the port level. Logging activation confirms that the mechanism is operational, it establishes the point from which the enforcement of 802.1X is in effect.
Condition	Requires 802.1X to be deployed and configured.

event_id	SEC-006
type	Security Mechanism State
action	enable
object_type	security_control
object_id	<interface_id> or <device>
event_result	state_change
event_criticality	medium

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.

6.11.7 802.1X disabled

Description	802.1X port-based network access control was deactivated on the device/on a specific interface.

Rationale	Disabling 802.1X removes authenticated access control at the port level, allowing any device to connect without verification. In OT environments this weakens the network boundary. Logging deactivation of 802.1X is essential for detecting removal of the control.
Condition	Requires 802.1X to be deployed and configured.

event_id	SEC-007
type	Security Mechanism State
action	disable
object_type	security_control
object_id	<interface_id> or <device>
event_result	state_change
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	—
Message	Not covered in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	—
Message	Not observed as a distinct event.

6.11.8 Key Management activities

Description	A cryptographic key or certificate was installed, renewed, deleted, or otherwise modified on the device.
Rationale	Key management changes affect the integrity of encrypted management channels, IPsec tunnels, and 802.1X authentication. Logging these events supports detection of unauthorized certificate replacement or key compromise response.
Condition	Requires device to support certificate/key management and to generate syslog events for key lifecycle operations.

event_id	SEC-008
type	Key Management
action	modify
object_type	security_control
object_id	<certificate _id> or <key_type>
event_result	success
event_criticality	high

Examples:

Syslog Security Events - SCALANCE & RUGGEDCOM Network components [9]	
Model	RUGGEDCOM RX1400
Message	FAILURE. The firmware integrity check has failed.
See also	Section 2.17.1, Table 2-78
Note	Partial coverage. The integrity check failure event is related to key/certificate validation but is not a key management lifecycle event. Dedicated key install/renew/delete events are not present in [9].

Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10 [10]	
Event ID	3420
Message	Web server certificate migrated successfully.
See also	Events 3421, 3422, 3423, 3424, 3425, 3426, 4759, 4760, 4761, 4762, 4763, 4764, 4769, 5552, 5553
Note	Extensive coverage. Crypto Events chapter covers certificate migration (3420–3422), CSR generation (4759), self-signed cert creation (4760), CRL download (4762–4764), certificate revocation (4761), and SSH rekey (5552, 5553).

Sources

[1] Syslog RFC 5424

[2] Syslog RFC 5425

[3] Syslog RFC 6587

[4] Syslog RFC 5426

[5] IEC 62443 (2-1, 2-4, 3-3, 4-2)

[6] ISO/IEC 27001:2022

[7] ISO/IEC 27002:2022

[8] ISO/IEC 27019:2024

[9] Syslog Security Events - SCALANCE & RUGGEDCOM Network components

[10] Aruba Event Log Messages Reference Guide for AOS-S Switch 16.10

[11] Planet switch log docs